TrES-2b

R-1687

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-2_190909` · created 2026-08-02T12:39:39+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458735.80173 +/- 0.00067 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.027
Transit depth (Rp/Rs)^2	2.22 +/- 0.81 [%]
Orbital Inclination (inc)	83.11 +/- 1.23
Airmass coefficient 1 (a1)	1.001 +/- 0.013
Airmass coefficient 2 (a2)	-0.0025 +/- 0.0059
Scatter in the residuals of the lightcurve fit is	1.27 %
Best Comparison Star	#1 - [459, 81]
Optimal Aperture	4.54
Optimal Annulus	8.13
Transit Duration (day)	0.059 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.027 vs 0.1254 (deviation 0.87σ)
Duration fit vs published	0.059 d vs 0.0746 d (deviation 0.56σ)
Mid-time O–C	-0.1 min
Depth SNR	5.5
Residual scatter	1.27 %

Light curve

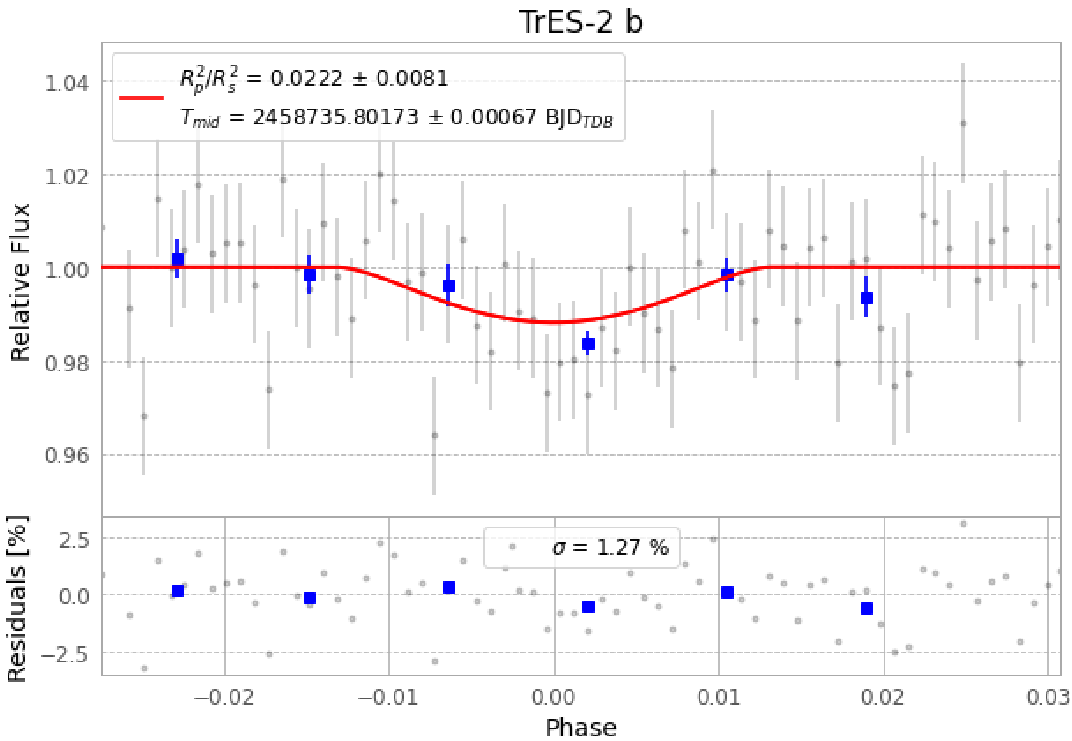

Figure 1 — FinalLightCurve_TrES-2 b_2019-09-08.png (SHA-256 ff17a856ecd374c0…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [362, 243], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 da1a2df7cc6c39bf…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

70 FITS frames (46 MB), `TRES-2190909053312.FITS` → `TRES-2190909090012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-190909.log (cc7509331921…), FinalLightCurve_TrES-2 b_2019-09-08.png (ff17a856ecd3…), FinalLightCurve_TrES-2 b_2019-09-08.csv (3d9a8eafd923…)

Compute resources

Wall time 200.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)